Higher National Diploma in Engineering (Civil)/ Higher National Diploma in Quantity Surveying

1st Year, 2nd Semester Examination – 2018

CE 1206/QS 12022– Construction Materials

Instructions to the candidates:
Answer any four (04) questions
All questions carry equal marks.

No of questions: 05
No of pages : 03
Time: **Two (02) hours**

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(01) Most of the Construction Materials are obtained from natural resources.

I. Classify rocks based on the process of their formation. (03 Marks)

II. List the total process of aggregate production from extraction to storing. (05 Marks)

III. List down the major types of crushers used in the aggregate production. (05 Marks)

IV. "Granite is a desired building material". Briefly explain this statement. (03 Marks)

V. Aggregates can be classified in many ways. Classify aggregates based on their grain size. (02 Marks)

VI. What are the desirable properties of aggregates? (03 Marks)

VII. Write down the name of any four experiments usually carried out to determine the quality of aggregates. (04 Marks)

(02) The applications of cement over various fields of construction have made it a very important Civil Engineering material.

I. What is cement? (02 Marks)

II. What are the raw materials used in cement production? (03 Marks)

(05 Marks)

III. What are the properties of cement?

IV. What is the role of Calcium Sulphate (Gypsum) in the production of cement? (03 Marks)

V. Briefly explain the manufacturing process of ordinary Portland cement using appropriate sketches. (05 Marks)

VI. List down the four field and laboratory tests used to determine the quality of any cement. (05 Marks)

VII. What types of cement do you propose for construction in costal belt area? (02 Marks)

(03) Bricks are the most common material used for the construction of superstructures.

I. Classify bricks based upon the physical and mechanical properties. (04 Marks)

II. What are the raw materials used for making bricks? (04 Marks)

III. List any five qualities of good quality clay bricks. (05 Marks)

IV. List any five disadvantages of using clay bricks (05 Marks)

.V. There are four different operations involved in the process of manufacturing of bricks: What are they? (04 Marks)

VI. What are the field tests carried out on bricks? (03 Marks)

(04) Concrete is a universal material used in construction works.

I. Briefly explain the terms "Plain cement concrete" and "Reinforced cement concrete" . (04 Marks)

II. What do you understand by the term "light weight concrete"? What are the advantages and disadvantages of using it. (06 Marks)

III. Briefly explain the influences of accelerators and retarders in concrete production. (04 marks)

IV. What do you understand by the term "workability of concrete"? (02 Marks)

V. What are the tests carried out for measuring the workability of concrete? (03 Marks)

VI. Explain why compaction of concrete is important? (03 Marks)

VII. Explain the importance of curing of concrete. (03 Marks)

Q5. Timber is a heavily used material for roofs, doors and windows.

I. Classify trees based on their growth pattern. (05 Marks)

II. What is meant by seasoning of timber? (05 Marks)

III. What are the advantages of seasoned timbers? (05 Marks)

IV. What are the major properties of timber? (04 Marks)

V. What are the most common defects found in timbers? (04 Marks)

VI. What do you understand by "deterioration of timber"? (02 Marks)